Evaluation Sets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach case coverage and honest denominators.

Supply complete sources and constructed outputs, preserve exposure roles and assess new ambiguous/missing/conflicting records.

Keep one habit: Ask what was tested, what failed and which expected behaviors still lack cases.

MAIN LESSON · SECTION 1

Prepare the main lesson and its retained extension

> Use fictional records and schedule separate extension time.

Goal and preparation

Primary target: Explain the **mechanism** and terminology of system **evaluation set**s

Two 45-minute sessions. A system **evaluation set** contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can explain **evaluation set**s using a traced example and a stated limitation.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Main plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval and readiness

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

Look for: Judge an output using separate observable rubric criteria

Reference example, not the only acceptable wording: Apply source support, task fit and sentence limit separately with exact **evidence**. A failed mandatory support gate blocks release despite other passes; the small rubric does not establish every quality or learner benefit.

Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of conversation state and retention

Reference example, not the only acceptable wording: Teacher explanation: Conversation state stores facts or prior interactions for later use.

It needs scope, provenance, correction and retention rules rather than indefinite accumulation. This classroom assistant keeps an approved room ID only for the current session.

A later correction replaces the active value but retains an audit event until session end. Do not persist private text merely because it appeared in conversation.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ **Materials and extension planning**

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Apply the connected idea: Ranking and retrieval evaluation

Two 45-minute sessions. A ranking needs task-specific relevance judgments and a declared **cutoff**. Precision and recall have different denominators. Use the supplied constructed scenarios to make the **mechanism** inspectable.

Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Case exposure check: this source example appears in ai-3/25, ai-3/26. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Quality Rubrics

Block to Python Bridge

Conversation State and Memory

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A test set contains ten easy questions with direct answers in the notes. Add three different kinds of cases that probe fallback behaviour and define expected outcomes.

Concept to explain

A good eval set represents real use, edge cases, and known risks.

Term	Meaning
evaluation set	Cases used to measure a system under declared criteria; specify whether they are validation or final test cases.
coverage	The range of relevant cases, conditions or groups included in data or tests.
expected behavior	The result or action required by the specification for a given case, stated before observing the system.
regression test	A repeatable check that previously correct behaviour still works after a change.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach case-level behavior before aggregate rates

> Separate the six-case visual, ten-case practice and ranking extension.

State the case-record contract

Case record	What to freeze before scoring
Identity	Case ID, user question and requested time
Evidence	Eligible source IDs/versions and relevant passages
Expected behavior	Supported claim, clarification, fallback or access refusal
Rubric	Required conditions and mandatory failures
Exposure	Development/reviewed or reserved; who has seen it
Observed result	Actual supplied output or recorded run, distinct from prediction

Case record	What to freeze before scoring
Aggregation	Report subgroup numerator/denominator before the pooled rate

A system **evaluation set** contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

Trace ten-case subgroup rates

Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

Read the pictured six-case record

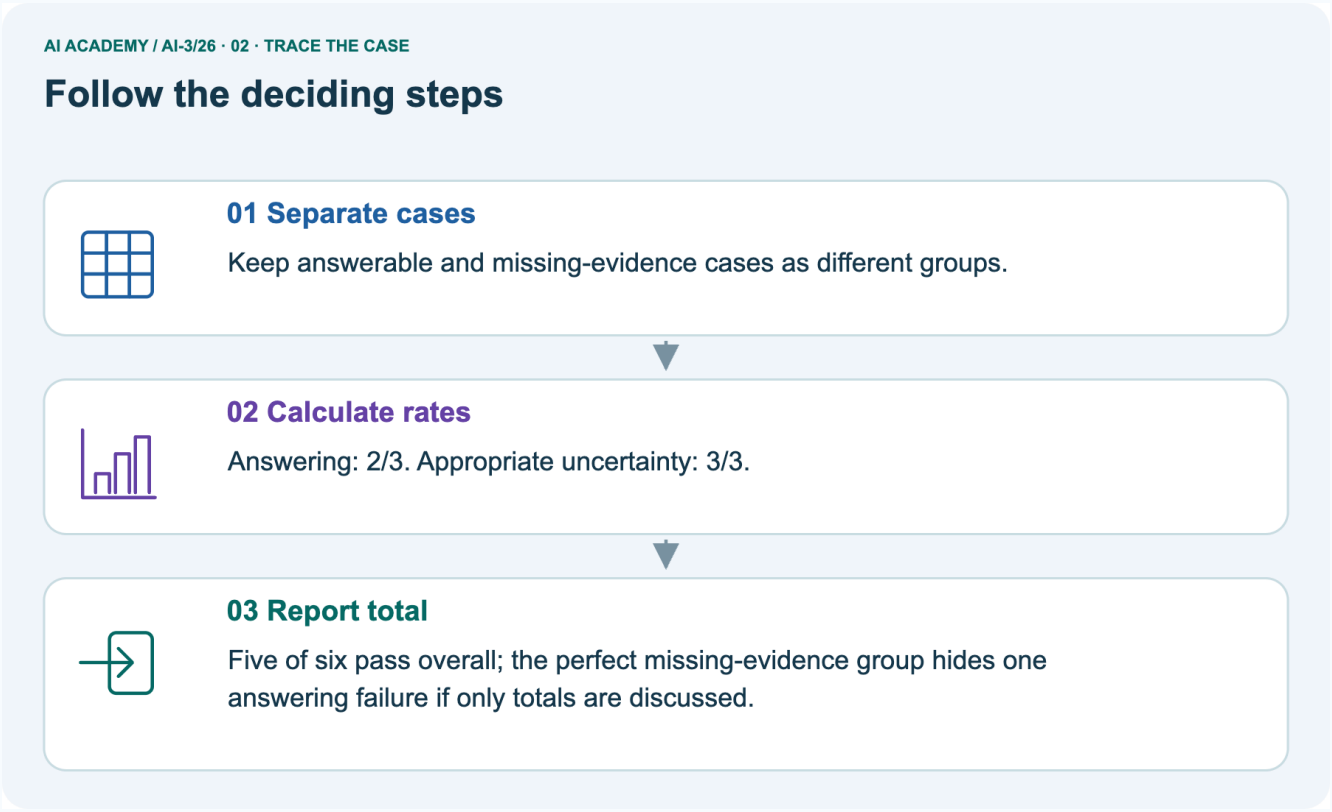
Design an **evaluation set** that checks answering and appropriate uncertainty separately.

Six toy questions: three have source answers and three do not. Results: 2 supported answers out of 3; 3 correct “not in source” responses out of 3.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep three records separate: the pictured six-case record has answering 2/3 and fallback 3/3, giving 5/6. Adding three successful missing-evidence cases gives 8/9 overall while answering remains 2/3. The separate ten-case W2 has answering 5/6 and fallback 3/4, giving 8/10. No system was run; these are supplied classroom outcomes, not estimated future accuracy.

Trace category and pooled rates



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Separate cases	Keep answerable and missing-evidence cases as different groups.
2. Calculate rates	Answering: 2/3. Appropriate uncertainty: 3/3.
3. Report total	Five of six pass overall; the perfect missing-evidence group hides one answering failure if only totals are discussed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict adding only fallback successes

Change: Add three missing-**evidence** cases that all pass.

Prediction: Overall becomes 8/9, but supported answering remains 2/3.

Reason: Changing the case mixture raises the pooled rate without improving any existing answerable result.

State coverage and generalisation limits

A small constructed set probes specific behaviours; its pooled rate is not an estimate of every future classroom question.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Extension · Judged ranking denominators

A ranking needs task-specific relevance judgments and a declared **cutoff**. Precision and recall have different denominators.

Precision at k is relevant returned items divided by k when exactly k items are returned. **Recall at k** is relevant returned items divided by all known relevant eligible items.

If the relevant set is unknown, do not invent recall.

Guided: Precision@3=2/3; recall@3=2/3. Duplicates should not count as distinct retrieved documents. Qualification: Known judgments may be incomplete; a score over one query does not establish **coverage** over all queries.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

▼ Additional practice · Easy definitions only

Ten evaluation questions are easy definitions; none lacks a source.

What is untested?

Absent **evidence**, ambiguity and boundary handling. Add predefined fallback cases and keep final cases separate from tuning.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Ten cases include factual, comparison, calculation, ambiguous, and missing-source questions.

2. Add conflicting-source, prompt-injection, sensitive-data, and accessibility cases.

3. Write **expected behavior** before running the product.

4. After a repair, keep the failed case as a **regression test**.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Evaluation case	An input with evidence and prespecified expected behaviour
Coverage	Which relevant conditions are represented

Term	Meaning
Contract pass	Satisfying the declared requirements for one case

Apply the connected idea: Ranking and retrieval evaluation

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

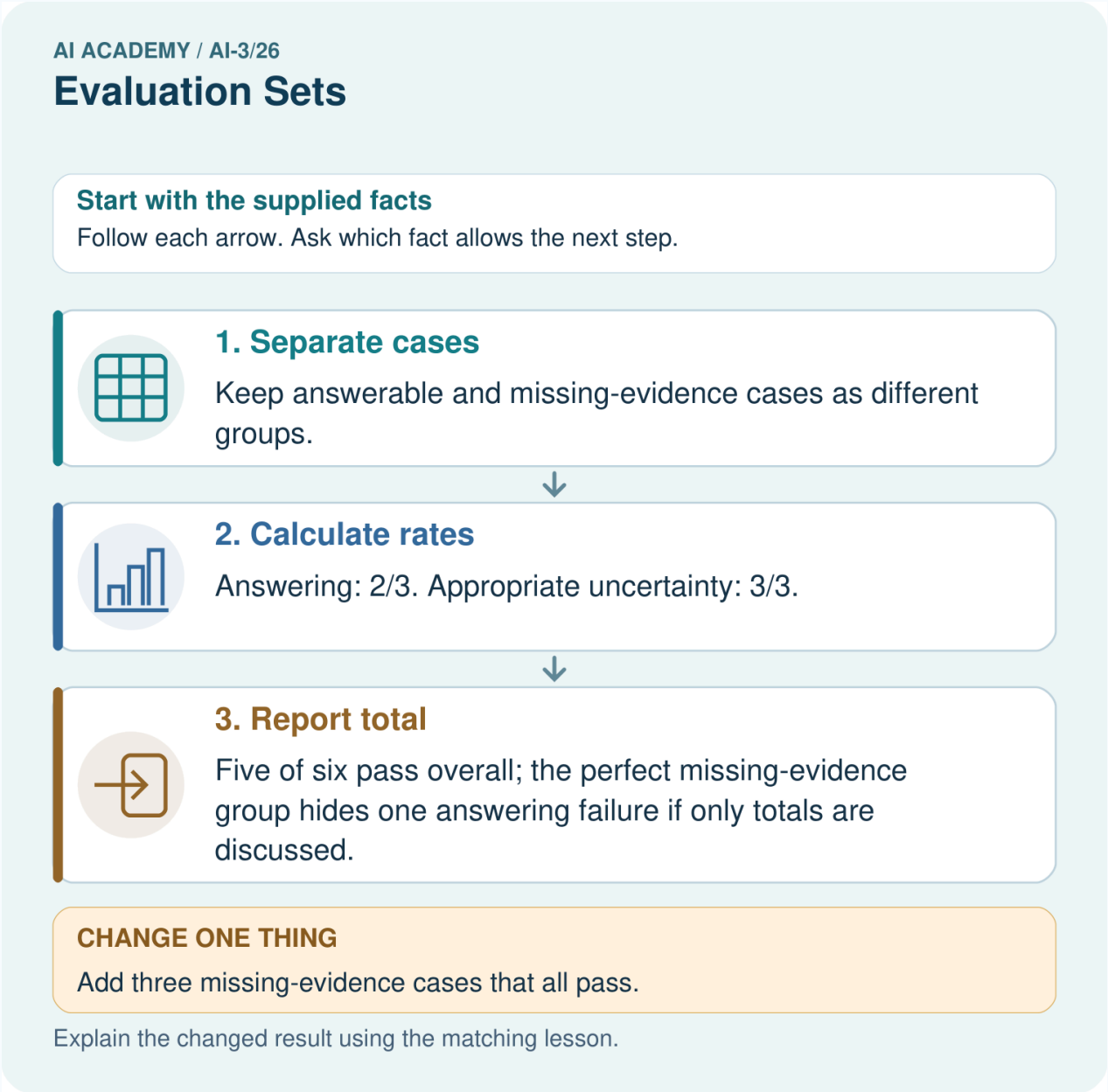
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Evaluation Sets

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the main explanation, then fade support

> Collect each learner’s reasoning before revealing a key.

Main completion question and key

Completion task: Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Main individual check

Use the worked case for System **evaluation set**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed case

Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No. Answerable cases still pass 2/3. Report both group rates and the changed evaluation mixture. Changed-case check: Overall becomes 8/9, but supported answering remains 2/3.

Changing the case mixture raises the pooled rate without improving any existing answerable result.

Changed-case reasoning: Overall becomes 8/9, but supported answering remains 2/3. Changing the case mixture raises the pooled rate without improving any existing answerable result.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep learners explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Extension teaching cycle

Target: Explain the **mechanism** and terminology of ranking and retrieval evaluation

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Relevant documents are A, C, E. Top three results A, B, C contain two relevant items. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Precision@3=2/3; recall@3=2/3. Duplicates should not count as distinct retrieved documents.

Individual check: For Ranking and retrieval evaluation, use its worked case to evaluate this proposed explanation: “The worked result establishes every future case”. Explain the deciding step.

Expected correction: Known judgments may be incomplete; a score over one query does not establish **coverage** over all queries.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Ten evaluation questions are easy definitions; none lacks a source.

Write the question to explain: What is untested?

Reveal after **prediction**s: Absent **evidence**, ambiguity and boundary handling. Add predefined fallback cases and keep final cases separate from tuning.

Use the earlier transfer question as optional practice. Reserve Design checks for the fresh club records independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

Qualification: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Teach the connected idea

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

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32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Connected teaching cycle: Ranking and retrieval evaluation

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied main investigation

> Keep the selected independent task out of shared practice.

Supplied records and answer guidance

Ten practice cases: A4 wrong red box and U2 unsupported lunch fail; other 8 pass. Answering 5/6, fallback 3/4, pooled 8/10. Six-case picture remains 2/3answering when total rises 5/6 to 8/9 through added fallback successes. Independent club task: clarify which club, request missing drama-leader evidence, preserve Robotics Room 2/Room 3 conflict until source authority/version is resolved.

Ranking extension uses known relevant set and distinct top-k IDs; do not infer unknown recall.

Predict, investigate and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the complete ten-case H/R/C practice set and supplied constructed outputs. Freeze source versions and expectations, score case-level outcomes, report subgroup rates, then identify missing ambiguity/conflict/access **coverage**. Newly authored cases have predictions, not fabricated results.
4. What did the hardest test teach you that the normal test did not?

Keep the evidence

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Main misconceptions

The worked result establishes every future case: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

The worked result establishes every future case: Known judgments may be incomplete; a score over one query does not establish **coverage** over all queries.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied main investigation for shared practice. Reserve Design checks for the fresh club records independently for the individual first attempt. The named extension has separate tasks and feedback.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

Assess the selected main independent task

> Reveal its key after the first attempt; extension tasks have their own keys.

Selected independent question and key

Selected case: Approved notes: Robotics Tuesday Room2; Art Wednesday Room5. Requests: “Where does the club meet?” and “Who leads drama?” An unversioned added note says Robotics Room3. Design checks and safe responses.

Clarify which club for the ambiguous request; the leader is absent, so abstain or request a source. Flag the conflicting Robotics room and resolve source authority/version before answering it.

Include ambiguity, absent facts and conflicts in evaluation; easy answered cases alone do not establish robustness.

Judge reasoning and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A system **evaluation set** contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

W2 Trace the worked case

Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

W3 Audit the limit

A report omits this qualification: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Extension questions and keys

Ranking and retrieval evaluation — W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A ranking needs task-specific relevance judgments and a declared **cutoff**. Precision and recall have different denominators.

Precision at k is relevant returned items divided by k when exactly k items are returned. **Recall at k** is relevant returned items divided by all known relevant eligible items.

If the relevant set is unknown, do not invent recall.

Ranking and retrieval evaluation — W2 Trace the worked case

Relevant documents are A, C, E. Top three results A, B, C contain two relevant items. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Precision@3=2/3; recall@3=2/3. Duplicates should not count as distinct retrieved documents.

Ranking and retrieval evaluation — W3 Audit the limit

A report omits this qualification: Known judgments may be incomplete; a score over one query does not establish **coverage** over all queries.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Known judgments may be incomplete; a score over one query does not establish **coverage** over all queries.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

Ranking and retrieval evaluation — W4 Apply a new case independently

Relevant set A, D, F, G; top three A, B, D. Compute precision and recall. Explain which omitted records affect recall.

Teacher explanation: Precision@3=2/3; recall@3=2/4=1/2. F and G are relevant omissions. The metric uses this judged set only.

Ranking and retrieval evaluation — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Known judgments may be incomplete; a score over one query does not establish **coverage** over all queries.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Absent **evidence**, ambiguity and boundary handling. Add predefined fallback cases and keep final cases separate from tuning.

Additional transfer practice

You improve a prompt after reading every test question. Can those questions still provide an untouched final evaluation? Propose a repair.

Reasoned practice answer

They have become development examples. Keep a separate unseen **evaluation set** and report the development results honestly.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A test set stays independent after its answers guide revisions.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Add an absent fact requiring 'not enough **evidence**'; an ambiguous referent requiring clarification; and conflicting notes requiring an attributed conflict rather than an invented reconciliation.

Keep final cases separate from tuning and specify acceptable answers before runs.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Twelve cases: eight answerable, four unanswerable. Correct answers six; appropriate abstentions two. Compute separate and combined pass rates.

Teacher explanation: 6/8=75%; 2/4=50%; combined 8/12=2/3. Do not call all twelve answer-accuracy cases.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and return

> Keep main and extension assessment evidence distinguishable.

Project record

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Main support and fresh assessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Selected independent task: ask which club for the ambiguous request; drama leader is absent, so request a source or state unavailable. Preserve Robotics Room 2/Room 3 conflict and ask for version/authority before an unconditional room answer. Design expected checks, not fabricated observed outcomes. Fresh retry: two club names, absent coach record and conflicting unversioned venue notes; require clarification, explicit missing evidence and unresolved authority. Earlier twelve-case rates remain additional practice.

Delayed return

Suggested return points in this level: ai-3/27, ai-3/29, ai-3/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 27: Red-Team the Product. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 27: Red Team the Product. Carry forward frozen expectations, case-level failures and untested risk categories when constructing adversarial tests.

▼ Extension support and reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Precision@3=2/3; recall@3=2/4=1/2. F and G are relevant omissions. The metric uses this judged set only. Require intermediate reasoning and the stated scope. Fresh retry: Relevant set B, E; top four A, B, C, D.

Expected reasoning: Precision@4=1/4; recall@4=1/2. If no relevant items exist, report recall denominator zero under an explicit policy.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Maintain a balanced eval set covering utility, grounding, format, privacy, and safety.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Bridge to the next week

Support and fresh reassessment

Readiness and reassessment for Ranking and retrieval evaluation

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative teaching checklist**

REFERENCE · SECTION 7

Reference · Alternative teaching checklist

> Use as a reminder alongside the main route.

Planning reference

Teaching target: Explain the **mechanism** and terminology of system **evaluation set**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

Check every learner before discussion: Use the worked case for System **evaluation set**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add three missing-**evidence** cases that all pass. Require a **prediction** and a reason before comparing. Expected result: Overall becomes 8/9, but supported answering remains 2/3.

Reason: Changing the case mixture raises the pooled rate without improving any existing answerable result.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and retained evidence**

REFERENCE · SECTION 8

Reference · Feedback and retained evidence

> Keep supported practice separate from independent evidence.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

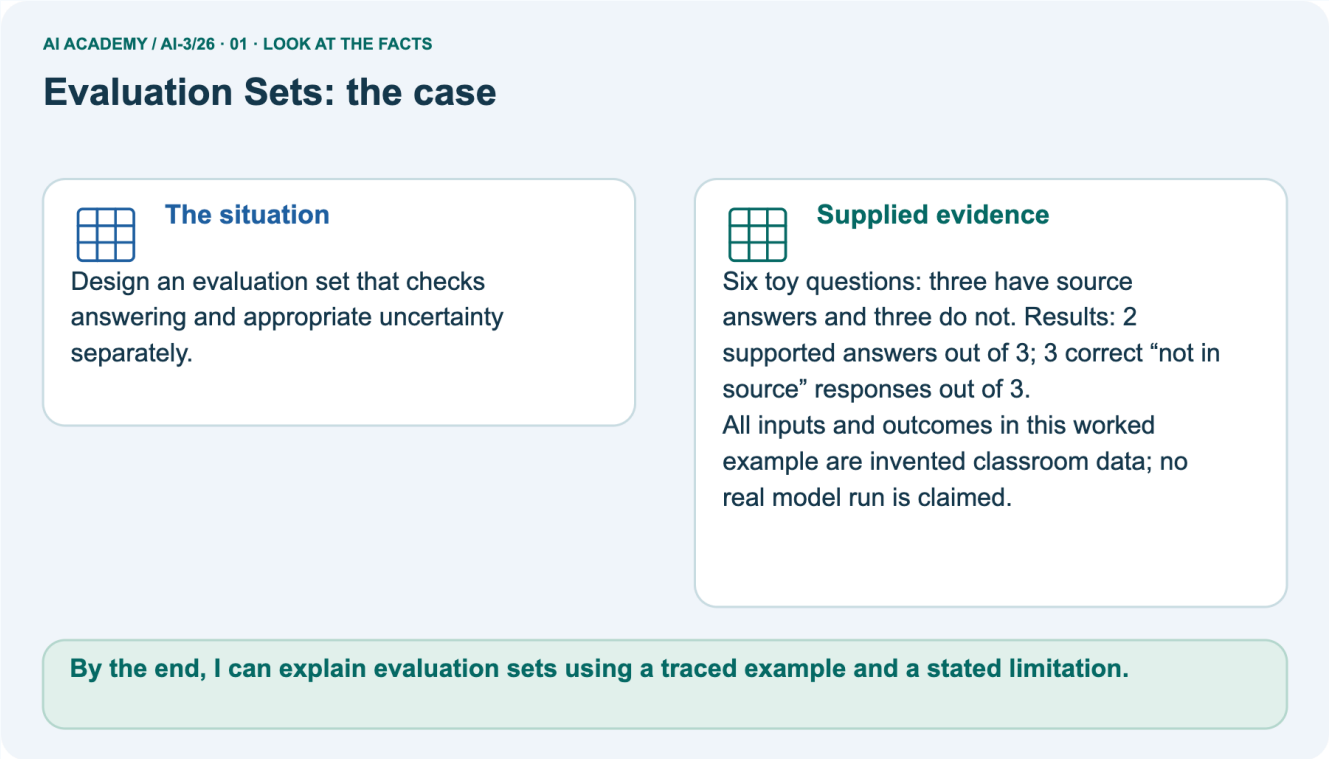
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose a picture for a specific explanation.

01. Evaluation Sets: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

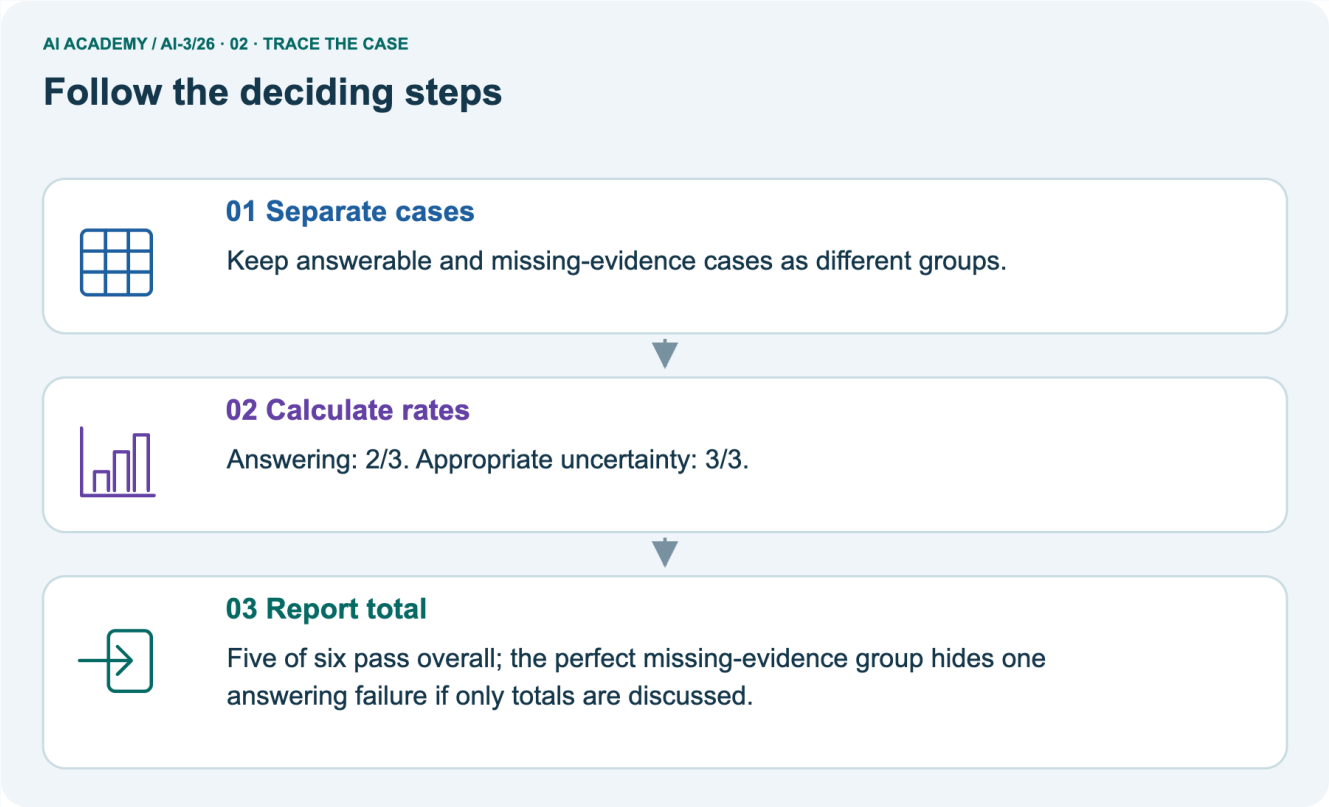
Design an **evaluation set** that checks answering and appropriate uncertainty separately. Supplied facts: Six toy questions: three have source answers and three do not.

Results: 2 supported answers out of 3; 3 correct “not in source” responses out of 3. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

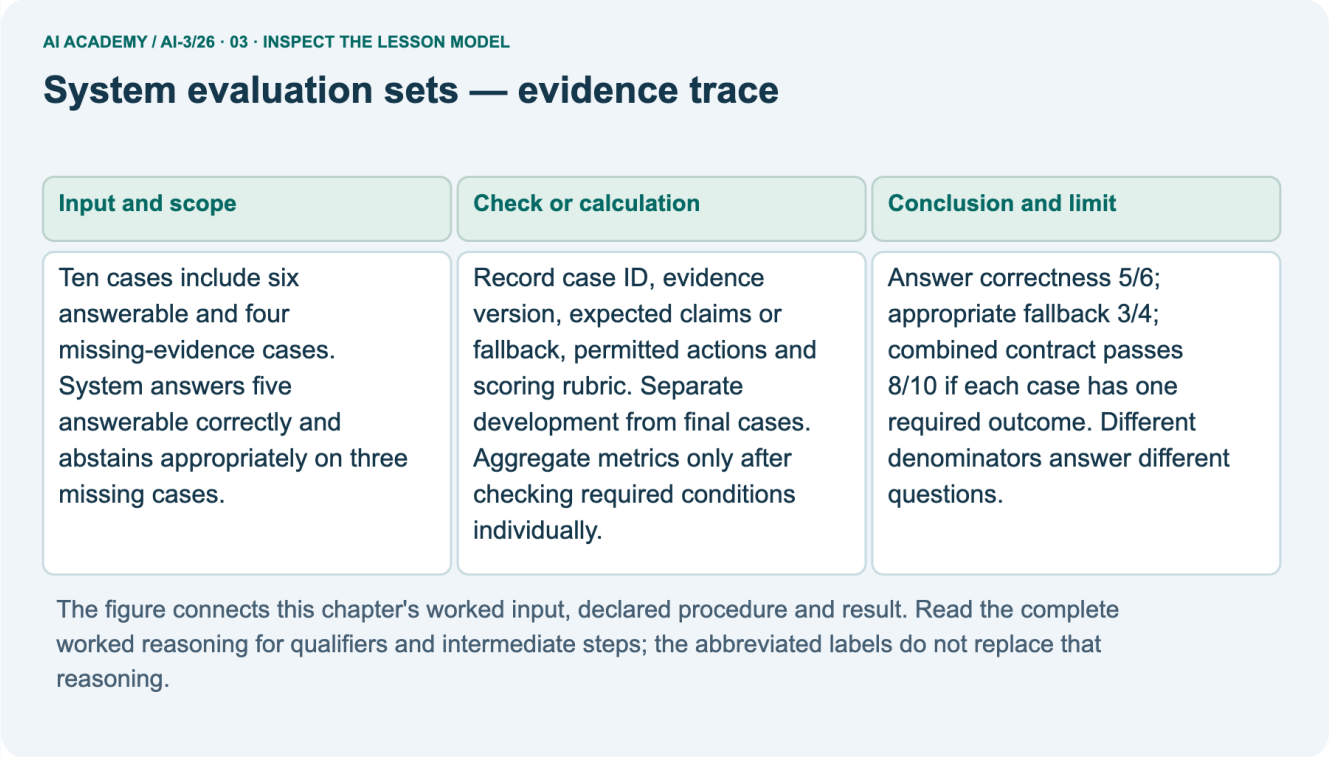
Separate cases: Keep answerable and missing-**evidence** cases as different groups. → Calculate rates: Answering: 2/3.

Appropriate uncertainty: 3/3. → Report total: Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. System evaluation sets — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

System **evaluation set**s — **evidence** trace. Ten cases include six answerable and four missing-**evidence** cases.

System answers five answerable correctly and abstains appropriately on three missing cases. | Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric.

Separate development from final cases.

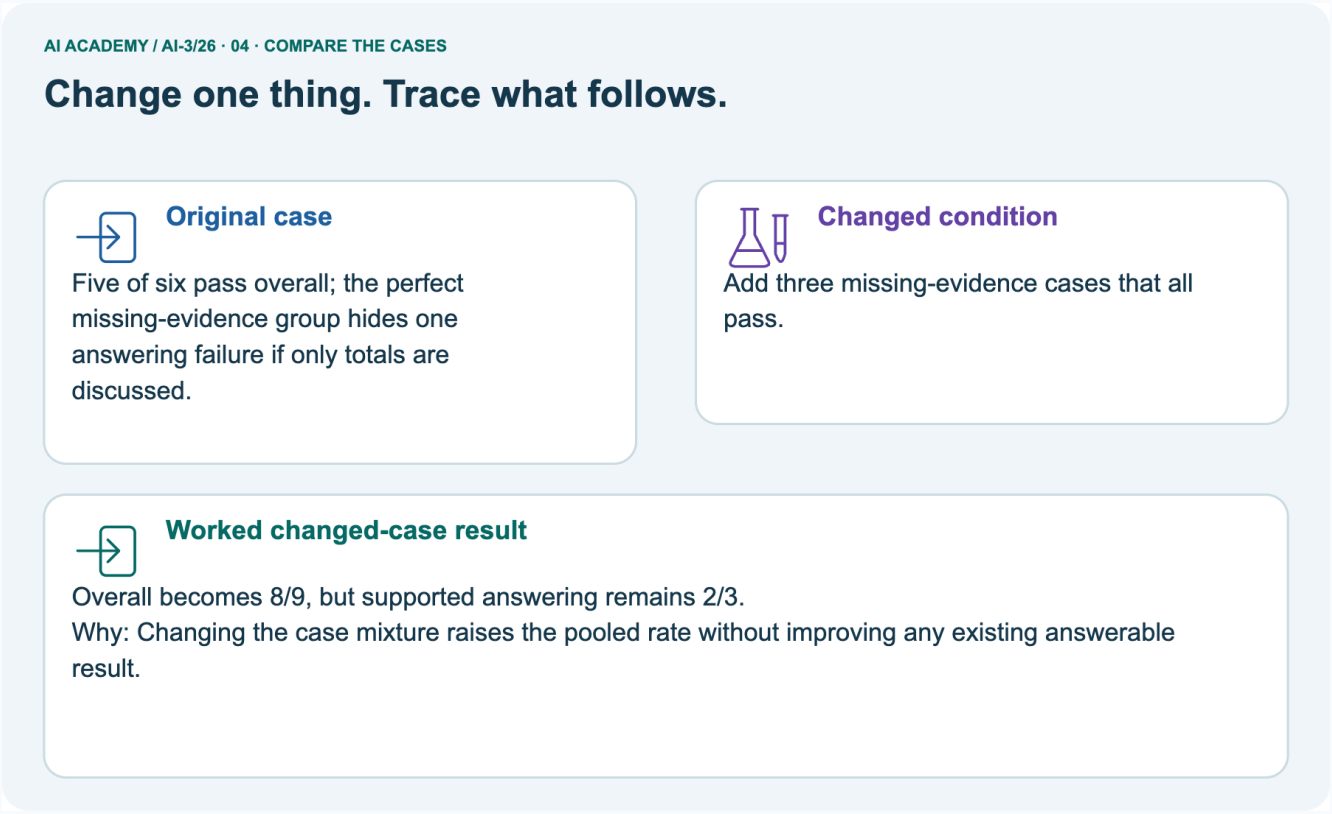
Aggregate metrics only after checking required conditions individually. | Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome.

Different denominators answer different questions.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. Changed condition: Add three missing-**evidence** cases that all pass.

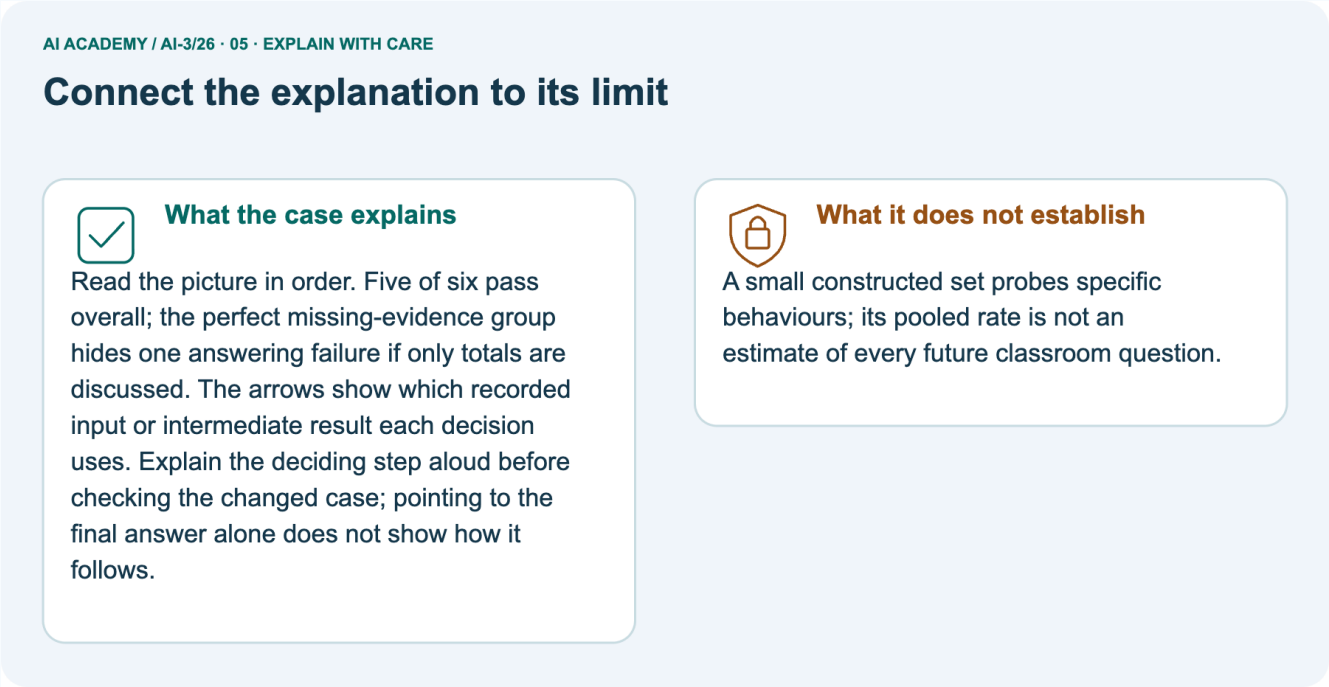
Expected result: Overall becomes 8/9, but supported answering remains 2/3. Reason: Changing the case mixture raises the pooled rate without improving any existing answerable result.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Changing the case mixture raises the pooled rate without improving any existing answerable result.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

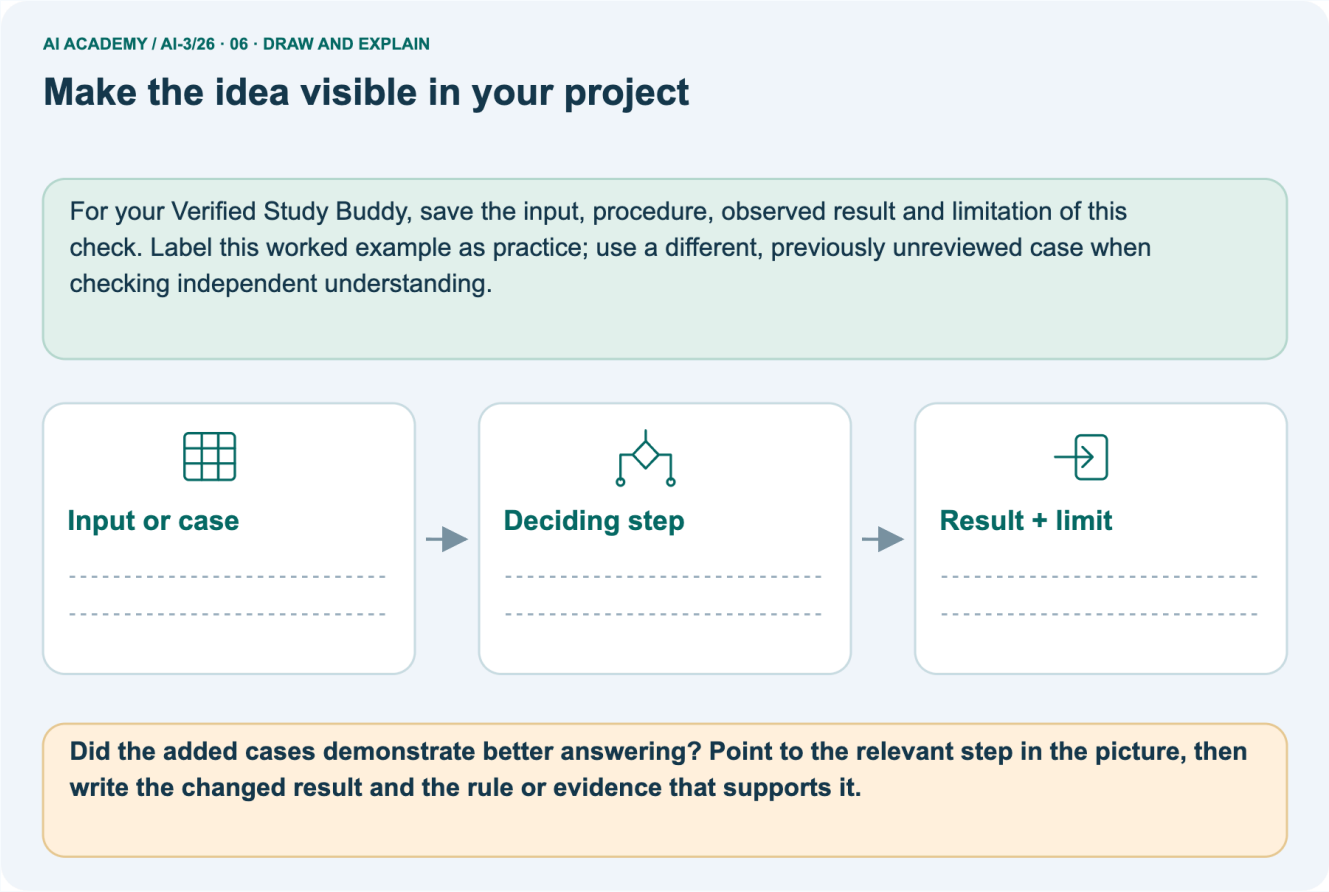
Limit: A small constructed set probes specific behaviours; its pooled rate is not an estimate of every future classroom question.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A small constructed set probes specific behaviours; its pooled rate is not an estimate of every future classroom question.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Answerable cases still pass 2/3. Report both group rates and the changed evaluation mixture. Changed-case check: Overall becomes 8/9, but supported answering remains 2/3.

Changing the case mixture raises the pooled rate without improving any existing answerable result.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Evaluation Sets: the case

02. Follow the deciding steps

03. System evaluation sets — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.